ETALE LOCAL CONSTRUCTIBILITY IS PRESERVED BY INVERSE IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a presentation reduction; the identification of the two preimages is the only computational step and it is stated explicitly.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Etale local constructibility tested through an etale cover. Properties of Spaces, Definition spaces-properties-definition-locally-constructible, used at proof lines 5932-5933. Is the definition used to reduce and to conclude.
- The scheme case for inverse images of constructible subsets. Morphisms, Lemma morphisms-lemma-inverse-image-constructible, used at proof lines 5929-5931. Is applied to U -> V.
- Existence of surjective etale presentations fitting into a square. used at proof lines 5925-5927. Produces V and U with U -> X surjective etale.
- Identification of the two preimages in U. used at proof lines 5928-5929. Turns the claim into the scheme statement.

Source reference. Original lemma 054I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6641–6641. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 0ECU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17969–17969. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $E \subset |Y|$ be a subset. If E is étale locally constructible in Y, then $f^{-1}(E)$ is étale locally constructible in X.*

Proof. Choose a scheme V and a surjective étale morphism $\varphi : V \to Y$. Choose a scheme U and a surjective étale morphism $U \to V \times_Y X$. Then $U \to X$ is surjective étale and the inverse image of $f^{-1}(E)$ in U is the inverse image of $\varphi^{-1}(E)$ by $U \to V$. Thus the lemma follows from the case of schemes for $U \to V$ (Morphisms, Lemma 054I) and the definition (Properties of Spaces, Definition 0ECU). □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.